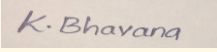


SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
CO5 ASSESSMENT TOOL 1 – Research Paper Review / Literature Survey

Course Code:	CSA1008	Course Title:	Software Engineering
CO Assessed:	CO5	Assessment:	Assessment Tool 1 – Research Paper Review / Literature Survey
Weightage:	20%	Total Marks:	25
CO5 Statement:	Evaluate emerging technologies and societal impacts, maintaining and evolving software systems responsibly		
Student Name:	K.BHAVANA	Reg. No.:	192524178
Date:	22-08-2026	Duration:	60 minutes

Code of Conduct

I K.BHAVANA(192524178)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action. Signature of the Student: 

Annexure A – Research Paper Review / Literature Survey

Instructions to Students:

Each student is assigned an individual problem statement. Based on the assigned problem statement, **conduct a literature survey and review relevant research papers** to understand the existing approaches, techniques, technologies, limitations, and research gaps related to the problem. The literature survey should include:

1. Identify and select **relevant research papers** related to the assigned problem statement from reputed journals, conferences, or scholarly databases.
2. Summarize the **objectives, methodologies, technologies, datasets, and major findings** of the selected research papers.
3. Compare the existing approaches based on suitable parameters such as **accuracy, performance, scalability, efficiency, methodology, or other problem-specific metrics**.
4. Identify the **limitations and challenges** of the existing research approaches.
5. Analyze the literature to identify the **research gap** or unresolved issues related to the assigned problem.
6. Discuss the **possible improvements or future research directions** based on the identified research gap.

7. Prepare a **comparative literature survey table** containing details such as author/year, research problem, methodology, dataset/tools, key findings, limitations, and research gap.
8. Provide proper **citations and references** using a consistent referencing style.

Deliverable: Submit a **Research Paper Review / Literature Survey Report** containing the selected research papers, individual paper reviews, comparative analysis, identified research gaps, future directions, and properly formatted references.

Case Study 33 – SDG 5:

Women Entrepreneurship Support Portal

Many women-owned small businesses struggle to access government schemes, financial assistance, and business mentoring. Existing information is fragmented across multiple websites. A government agency plans to create an integrated digital platform supporting registration, funding opportunities, training programs, and networking. As a software engineer, analyze the above problem and propose a suitable software engineering solution.

1. Introduction

Women entrepreneurs, particularly those running micro, small and medium enterprises (MSMEs), often find it difficult to locate and use government schemes, funding opportunities, training programmes and mentorship because this information is scattered across many separate government websites and offices. This fragmentation increases the time and effort needed for a woman entrepreneur to discover the support that is actually available to her, and it makes it harder for government agencies to track who is using their schemes.

This report is prepared as part of CO5 Assessment Tool 1 (Research Paper Review / Literature Survey) for the Software Engineering course (CSA10 / CSA1063), based on Case Study 33 under SDG 5 (Gender Equality). The assigned problem statement asks for a software engineering solution to this fragmentation problem. This report reviews existing academic research related to the problem and connects the findings to the Women Entrepreneurship Support Portal, a centralised digital platform proposed and previously documented as a project for this case study.

2. Case Study / Problem Statement

Case Study 33 describes the following situation: many women-owned small businesses struggle to access government schemes, financial assistance, and business mentoring because existing information is fragmented across multiple websites. A government agency plans to create an integrated digital platform supporting registration, funding opportunities, training programmes, and networking. As a software engineer, the task is to analyse this problem and propose a suitable software engineering solution.

The Women Entrepreneurship Support Portal, documented in a previous project report for this same case study, is the proposed response to this problem. It is a role-based web application that brings together scheme discovery, funding applications, training programmes, mentorship and networking activities for women entrepreneurs, government officers and mentors, built using React/Vite on the frontend and a Node.js/Express backend with a SQLite-based database, deployed with Docker.

3. Objective of the Literature Survey

The objective of this literature survey is to study existing academic research related to the problems addressed by the Women Entrepreneurship Support Portal, so that the design of the portal can be justified and improved using evidence from published studies. The survey focuses specifically on:

- Digital platforms and their role in supporting women entrepreneurs
- Government and e-government portals, including how their design affects citizen satisfaction
- Access to financial assistance and funding through digital channels
- Entrepreneurship training delivered through digital or app-based platforms
- Online mentoring and networking for entrepreneurs
- Service quality and trust in digital platforms used by women entrepreneurs

Papers that discuss gender equality only in a general sense, without a direct connection to entrepreneurship, digital platforms, business support, funding, training, mentoring, or government services, were excluded from this review.

4. Research Paper Selection

Seven research papers were selected for this survey. The papers were identified through Google Scholar and verified against their publisher records (Springer, Wiley, Elsevier/ScienceDirect, MDPI, and Oxford Academic/World Bank). Six of the seven papers were published between 2020 and 2025, and the seventh is a 2026 advance-access article that originally circulated as a 2024 working paper; all are peer-reviewed journal publications. Each paper was checked individually for its title, authors, year, publishing journal and DOI before being included.

No.	Paper Title	Author(s)	Year	Source	Relevance to Case Study 33
1	A Systematic Literature Review on Existing Digital Government Architectures	Baheer, Lamas & Sousa	2020	Administrative Sciences (MDPI)	Architecture for centralised one-stop government portals
2	Service Design and Citizen Satisfaction with E-Government Services	Chan, Thong, Brown & Venkatesh	2021	Public Administration Review (Wiley)	Design features that drive satisfaction with scheme/dashboard portals
3	Digital platforms and entrepreneurial support: a field experiment in online mentoring	Lall, Chen & Mason	2022	Small Business Economics (Springer)	Informs the mentor-directory / mentorship module
4	Can digital financial inclusion promote female entrepreneurship? Evidence and mechanisms	Yang, Huang & Gao	2022	North American Journal of Economics and Finance (Elsevier)	Supports the funding-opportunities / application module
5	The Role of Digital Platforms in Women's Entrepreneurial Opportunity Process: Does Online Social Capital Matter?	Hammad & El Naggat	2023	Human Behavior and Emerging Technologies (Wiley)	Supports the networking / community-activities module
6	Testing the Promise of Digital Scaling: In-Person versus App-Based Training for Women Entrepreneurs	Cassidy, Ebrahim & Ubfal	2026 (WP 2024)	World Bank Economic Review (Oxford Academic)	Directly informs the training-programme module
7	Performance of women entrepreneurs in digital platforms: a SERVQUAL approach	Roy Dutta	2025	Decision (Springer)	Informs service quality, trust and UX for the portal

5. Review of Research Papers

5.1 Paper 1

Title: A Systematic Literature Review on Existing Digital Government Architectures: State-of-the-Art, Challenges, and Prospects

Authors: Baheer, B. A., Lamas, D., & Sousa, S.

Year: 2020

Journal/Conference: Administrative Sciences (MDPI), Vol. 10, No. 2, Article 25

1. Research Problem

There is no single agreed architecture for building Digital Government infrastructure, even though many governments have tried to build centralised 'one-stop' portals that bring different government services into one place.

2. Objective

The authors wanted to systematically review existing Digital Government architectures to identify their common characteristics and building blocks, and to find out what gaps exist in this area of research.

3. Methodology / Approach

A systematic literature review was carried out. The authors collected 103 papers published between 2003 and 2020 from five major academic databases, and analysed them using a structured set of research questions and inclusion/exclusion criteria.

4. Technologies / Tools

The paper discusses architectural concepts such as Service-Oriented Architecture (SOA), Enterprise Architecture (EA), service registries, service discovery components, and one-stop portals, rather than any single named software tool.

5. Dataset

Dataset was not specified in the paper, as this is a literature review of 103 published studies rather than an empirical dataset.

6. Main Findings

The review found that Digital Government portals rely on common building blocks, including a service registry (to publish available services), a service discovery mechanism (to search for the right service), and a one-stop portal (a single access point for citizens and organisations).

However, there is no widely agreed reference architecture, so most systems are built in different, non-standard ways.

7. Limitations

The review is conceptual and does not test any architecture in a live system. It also notes that many existing studies stop at proposing an architecture rather than evaluating it in practice.

8. Relevance to Women Entrepreneurship Support Portal

This paper is directly relevant to the core idea of the Women Entrepreneurship Support Portal, which is to bring fragmented government scheme information into a single centralised platform. The building blocks identified in this review (a registry-style approach to schemes, and a single portal for discovery) support the architectural direction already taken in the existing project, where scheme discovery, eligibility checking and applications are handled through one integrated system rather than several disconnected ones.

5.2 Paper 2

Title: Service Design and Citizen Satisfaction with E-Government Services: A Multidimensional Perspective

Authors: Chan, F. K. Y., Thong, J. Y. L., Brown, S. A., & Venkatesh, V.

Year: 2021

Journal/Conference: Public Administration Review, Vol. 81, No. 5, pp. 874-894 (Wiley)

1. Research Problem

Even though many e-government services exist, it is not well understood which specific design features of these services actually lead to citizen satisfaction.

2. Objective

The study aimed to define and test a multidimensional model of e-government service design, identifying which design characteristics of core, facilitating and supporting services affect perceived service quality and, in turn, citizen satisfaction.

3. Methodology / Approach

A two-stage survey was carried out with 3,065 users of three real e-government services. The data was analysed using structural equation modelling to test the relationships between design characteristics, perceived service quality, and satisfaction.

4. Technologies / Tools

The paper studies real, deployed e-government service websites rather than a specific software tool; it uses survey and structural equation modelling techniques as its research method.

5. Dataset

The dataset was a two-stage survey of 3,065 users of three government e-services.

6. Main Findings

Design characteristics such as accuracy, completeness, self-service capability, convenience, accessibility, privacy protection, user support, personalisation and transparency all significantly contributed to perceived service quality, which in turn increased citizen satisfaction. The three types of service perception (core, facilitating and supporting) also interacted with and reinforced each other.

7. Limitations

The findings come from three specific e-government services, so results may not directly generalise to every type of government digital service.

8. Relevance to Women Entrepreneurship Support Portal

This paper gives concrete, tested design guidance for the government-facing parts of the Women Entrepreneurship Support Portal, such as the scheme discovery, eligibility-checking, and dashboard/reporting features. It supports building in accuracy of scheme information, self-service application capability, personalisation (showing schemes relevant to a specific entrepreneur's profile), and transparency (visible application status), since these are the design characteristics shown to raise citizen satisfaction with government digital services.

5.3 Paper 3

Title: Digital platforms and entrepreneurial support: a field experiment in online mentoring

Authors: Lall, S. A., Chen, L.-W., & Mason, D. P.

Year: 2022

Journal/Conference: Small Business Economics (Springer)

1. Research Problem

Mentorship is known to help entrepreneurs, but little is known about how entrepreneurs actually connect with mentors when this happens through a digital platform rather than in person.

2. Objective

The researchers wanted to test whether specific interventions on an online mentoring platform could increase the likelihood that entrepreneurs would reach out to, and successfully connect with, a mentor.

3. Methodology / Approach

A randomised field experiment was run on a real online platform that connects entrepreneurs with potential mentors. Entrepreneurs were randomly shown different interventions, including a video of a successful mentor-mentee relationship, and the researchers measured whether this changed the rate at which entrepreneurs initiated contact and formed a mentoring connection.

4. Technologies / Tools

An existing online mentor-matching platform was used to run the experiment and collect platform usage data.

5. Dataset

The dataset consisted of platform log data covering entrepreneurs' actions (such as initiating conversations and forming connections) on the mentoring platform.

6. Main Findings

Showing entrepreneurs a video of a successful mentor-mentee relationship increased the chance that they would reach out to a mentor, although it did not significantly increase the chance of successfully forming a connection. This effect was stronger for female entrepreneurs than for the overall sample.

7. Limitations

Not all entrepreneurs adopted the intervention, and reaching out to a mentor did not always lead to an established mentoring relationship, showing that outreach and successful connection are not the same outcome.

8. Relevance to Women Entrepreneurship Support Portal

This paper is directly useful for the mentor-directory and mentorship-workflow features already planned in the Women Entrepreneurship Support Portal. It suggests that simple platform design choices, such as showing success stories or examples of prior mentor-mentee interactions, can meaningfully increase how often women entrepreneurs actually reach out to mentors listed on the platform, rather than just browsing a directory.

5.4 Paper 4

Title: Can digital financial inclusion promote female entrepreneurship? Evidence and mechanisms

Authors: Yang, X., Huang, Y., & Gao, M.

Year: 2022

Journal/Conference: The North American Journal of Economics and Finance, Vol. 63, Article 101800 (Elsevier)

1. Research Problem

Women often face greater difficulty than men in accessing financing and business information, which makes it harder for them to start and grow a business.

2. Objective

The study aimed to examine whether digital financial inclusion (digital access to financial services and information) increases the likelihood that women start a business, and to understand the mechanisms through which this happens.

3. Methodology / Approach

The researchers matched a regional digital financial inclusion index with a nationally representative household survey and used statistical models, including a probit model, to test the relationship between digital financial inclusion and women's decisions to start a business.

4. Technologies / Tools

The study relies on an existing digital financial inclusion index rather than a specific software system; it is an empirical economics study using secondary data.

5. Dataset

Dataset was not specified in the paper beyond stating that it used a matched national household survey and a regional digital financial inclusion index.

6. Main Findings

Higher regional digital financial inclusion significantly increased the probability that a woman would start a business. The study found that digital financial inclusion works by easing women's financing constraints, improving their access to business information, and increasing their work flexibility.

7. Limitations

The relationship is measured at a regional level and is correlational, so it cannot fully separate digital financial inclusion from other regional factors that might also affect women's entrepreneurship.

8. Relevance to Women Entrepreneurship Support Portal

This paper supports the design of the funding-opportunities and funding-application modules in the portal. It suggests that giving women entrepreneurs digital, direct access to financial information and application channels, rather than requiring them to visit physical offices, is likely to genuinely increase their participation in entrepreneurship, which justifies making funding discovery and application a core, easily accessible feature of the platform.

5.5 Paper 5

Title: The Role of Digital Platforms in Women's Entrepreneurial Opportunity Process: Does Online Social Capital Matter?

Authors: Hammad, R., & El Naggar, R.

Year: 2023

Journal/Conference: Human Behavior and Emerging Technologies, Vol. 2023, Article 5357335 (Wiley)

1. Research Problem

It is not well understood whether the social capital that nascent women entrepreneurs build online, for example through social media communities, actually helps them recognise and act on entrepreneurial opportunities.

2. Objective

The study aimed to investigate whether online social capital, built through digital platforms, helps nascent female entrepreneurs develop entrepreneurial capacity and access the resources needed to recognise and exploit business opportunities.

3. Methodology / Approach

A mixed-method approach was used in two phases. In the first phase, the researchers used Natural Language Processing to analyse large-scale text data from a Facebook community of entrepreneurs. In the second phase, they built and tested a structural model using SmartPLS with survey data.

4. Technologies / Tools

Natural Language Processing (NLP) for social media text analysis, and SmartPLS 3.3 for quantitative structural equation modelling.

5. Dataset

The dataset consisted of big data collected from a Facebook community of entrepreneurs, combined with survey responses from nascent female entrepreneurs.

6. Main Findings

Nascent female entrepreneurs who build online social capital, particularly 'bridging' social capital across different social groups, are better able to develop their entrepreneurial capacity and access the resources needed to recognise and use business opportunities.

7. Limitations

The findings are based on one national context and one social media platform, so the results may not directly apply to other countries or other types of digital communities.

8. Relevance to Women Entrepreneurship Support Portal

This paper supports the inclusion of networking and community-activity features in the portal. It suggests that giving women entrepreneurs a structured space to build connections with other entrepreneurs, similar to the bridging social capital described in the paper, can genuinely help them recognise and access business opportunities, not just provide a social feature with no practical value.

5.6 Paper 6

Title: Testing the Promise of Digital Scaling: In-Person versus App-Based Training for Women Entrepreneurs

Authors: Cassidy, R., Ebrahim, M., & Ubfal, D.

Year: 2026 (advance article; originally circulated as a World Bank working paper in 2024)

Journal/Conference: The World Bank Economic Review (Oxford Academic)

1. Research Problem

Business training for entrepreneurs is normally delivered in person at high cost, and it is unclear whether delivering the same training through a smartphone app can achieve similar results at lower cost, particularly for women who may have time and mobility constraints.

2. Objective

The study aimed to test whether an app-based training programme performs as well as an in-person training programme with an identical curriculum, in terms of completion rates and business outcomes for women entrepreneurs.

3. Methodology / Approach

A randomised controlled trial (RCT) was conducted with 2,000 women entrepreneurs in Ethiopia. Participants were randomly assigned to an app-based training group, an in-person training group, or a control group, and outcomes were compared between groups.

4. Technologies / Tools

A purpose-built smartphone training application, compared against traditional in-person classroom-style training sessions.

5. Dataset

The dataset consisted of survey and administrative data collected from the 2,000 women entrepreneurs who took part in the randomised controlled trial in Ethiopia.

6. Main Findings

Both training formats had high initial take-up (over 75 percent), but completion rates were very different: only 22 percent for the app-based training compared with 71 percent for in-person training. Both formats produced only small, statistically limited improvements in business knowledge and practices.

7. Limitations

The study was conducted in one country (Ethiopia) with a specific target group, and the authors note that low digital literacy and time constraints reduced engagement with the app; results may differ in other settings with different levels of digital access.

8. Relevance to Women Entrepreneurship Support Portal

This paper gives an important, realistic caution for the training-programme module of the Women Entrepreneurship Support Portal. It suggests that simply making training available online is not enough to guarantee that women entrepreneurs will complete it; the portal's training module should be designed with structured schedules, reminders, or blended elements, rather than assuming self-paced digital delivery alone will be sufficient.

5.7 Paper 7

Title: Performance of women entrepreneurs in digital platforms: a SERVQUAL approach

Authors: Roy Dutta, M.

Year: 2025

Journal/Conference: Decision, Vol. 52, pp. 101-116 (Springer)

1. Research Problem

It is not clear which specific service-quality factors determine whether customers are satisfied with the digital businesses run by women entrepreneurs, which in turn affects the entrepreneurs' success on digital platforms.

2. Objective

The study aimed to assess how women entrepreneurs adapt to and perform on digital platforms, using customer feedback to measure satisfaction across five recognised service-quality dimensions.

3. Methodology / Approach

A survey-based SERVQUAL gap analysis was carried out, covering 250 respondents (both customers and women entrepreneurs) across rural and urban areas of West Bengal. Weighted average scores were calculated across five dimensions: Tangibility, Reliability, Responsiveness, Trust, and Empathy.

4. Technologies / Tools

The SERVQUAL service-quality framework, applied specifically to online/digital marketing by women entrepreneurs.

5. Dataset

The dataset was a survey of 250 respondents (customers and women entrepreneurs) covering rural and urban areas of West Bengal.

6. Main Findings

Overall gap scores between customer expectations and perceived service quality were negative, but the dimensions of Responsiveness, Trust and Empathy showed positive weighted scores, suggesting these are relative strengths that support the success of women-led digital businesses.

7. Limitations

The study is limited to one Indian state (West Bengal) and relies on self-reported survey data from a moderate sample size of 250 respondents.

8. Relevance to Women Entrepreneurship Support Portal

This paper is useful for the general quality and trust aspects of the Women Entrepreneurship Support Portal, beyond just the government-facing modules. It suggests that responsiveness (timely responses to funding or mentorship applications), trust (transparent verification of businesses and mentors), and empathy in how the platform communicates with entrepreneurs are likely to matter for how well the portal is received by its users.

6. Comparative Literature Analysis

6.1 Comparative Table

No.	Author/Year	Research Problem	Methodology	Technology/Tools	Dataset	Main Finding	Limitation
1	Baheer, Lamas & Sousa (2020)	No single reference architecture for centralised one-stop government portals	Systematic review of 103 papers (2003-2020)	SOA / EA architecture patterns	Not applicable (review)	Common building blocks: service registry, discovery, one-stop portal	No agreed reference architecture; mostly conceptual
2	Chan, Thong, Brown & Venkatesh (2021)	Unclear which design features drive citizen satisfaction	Two-stage survey; structural equation modelling	Not applicable (survey study)	3,065 users of 3 e-govt services	Accuracy, self-service, personalisation, transparency raise satisfaction	Context-specific to services studied
3	Lall, Chen & Mason (2022)	Entrepreneurs struggle to initiate mentor connections online	Randomised field experiment on live platform	Online mentor-matching platform + video nudges	Platform log data	Success-story videos increase mentor outreach, more so for women	Outreach did not always lead to a completed connection
4	Yang, Huang & Gao (2022)	Women face greater financing/information barriers to starting a business	Quantitative analysis; probit modelling	Digital financial inclusion index	National household survey matched with index	Higher digital financial inclusion raises probability women start a business	Regional-level, correlational evidence
5	Hammad & El Naggar (2023)	Unclear if online social capital helps women recognise opportunities	Mixed-method: NLP + SmartPLS modelling	NLP on social media text; SmartPLS 3.3	Facebook entrepreneur community data + survey	Bridging online social capital builds entrepreneurial capacity	Single country and single platform context
6	Cassidy, Ebrahim & Ubfal (2026)	Unclear if app-based training matches in-person training for women	Randomised controlled trial, 3 arms	Purpose-built training app vs in-person sessions	2,000 women entrepreneurs (Ethiopia RCT)	App completion (22%) far below in-person (71%); modest outcome gains either way	Low sustained engagement with app-only delivery
7	Roy Dutta (2025)	Unclear which service-quality factors drive satisfaction with women-run digital businesses	SERVQUAL survey gap analysis	SERVQUAL framework	250 respondents, West Bengal	Responsiveness, trust and empathy scored positively despite overall negative gap	Single-state sample; self-reported data

6.2 Comparison of Research Approaches

The seven papers use a mix of research approaches. Two papers (Baheer, Lamas & Sousa; and the comparative review itself) rely on secondary review of existing literature, while the

remaining five are primary empirical studies using surveys (Chan et al.; Roy Dutta), a randomised field experiment (Lall, Chen & Mason), a randomised controlled trial (Cassidy, Ebrahim & Ubfal), and a mixed NLP-plus-survey design (Hammad & El Nagggar). This mix is useful for the Women Entrepreneurship Support Portal because it combines architectural guidance (what to build) with behavioural evidence (how design choices affect real user behaviour).

6.3 Comparison of Technologies

None of the seven papers proposes or evaluates a single unified software platform combining schemes, funding, training, mentoring and networking in one system. Instead, each paper studies one piece of the puzzle in isolation: government portal architecture (Paper 1), e-government service design (Paper 2), a standalone mentoring platform (Paper 3), a financial inclusion index (Paper 4), a social media community (Paper 5), a standalone training app (Paper 6), and general digital marketing by women entrepreneurs (Paper 7). This confirms that the fragmentation described in Case Study 33 is reflected in the research literature itself, where each support function is usually studied and built separately rather than as one integrated system.

6.4 Comparison of Findings

A common thread across the papers is that digital access genuinely helps women entrepreneurs, but only when the digital service is well designed. Papers 2, 3, 4 and 5 all report positive effects of digital platforms on satisfaction, mentor outreach, business start-up, and opportunity recognition respectively. However, Paper 6 shows that digital delivery alone is not sufficient for training, since completion dropped sharply without in-person support, and Paper 7 shows that even where digital platforms are used, service-quality gaps remain. Together, the papers suggest that a digital platform can be highly effective, but only if it is designed around measurable service-quality factors rather than simply being made available online.

6.5 Comparison of Limitations

A common limitation across the papers is limited geographic or contextual generalisability: each study is based on a specific country, region, or single platform (for example, three e-government services in Paper 2, one Facebook community in Paper 5, or one Indian state in Paper 7). Several papers also rely on self-reported survey data (Papers 2 and 7), which can be less reliable than directly observed platform behaviour. Paper 1 highlights a limitation that

applies to the whole set of papers: most existing research proposes or studies individual components rather than evaluating a complete, integrated support platform for women entrepreneurs.

7. Critical Analysis

The reviewed literature shows that existing digital systems do several things well. Government e-service portals, when properly designed, are shown to be capable of achieving high citizen satisfaction through accuracy, self-service capability, personalisation and transparency (Paper 2). Online platforms are shown to genuinely help entrepreneurs form mentoring relationships (Paper 3), and digital access to financial services is shown to increase women's likelihood of starting a business (Paper 4). Online communities are shown to build the kind of social capital that helps women recognise business opportunities (Paper 5).

At the same time, the literature reveals problems that are consistently repeated. Systems tend to be fragmented rather than centralised: no paper in this survey studies a single platform that combines government schemes, funding, training and mentoring together, which mirrors the exact problem described in Case Study 33. Digital delivery is not automatically effective: Paper 6 shows that a training app on its own achieved far lower completion than in-person sessions, showing that digital availability does not guarantee real usage. Service-quality gaps persist even on digital platforms run by or for women entrepreneurs, as shown by the negative overall SERVQUAL gap scores in Paper 7.

Based only on what the seven reviewed papers support, the following can be said about common features: government scheme information is addressed through the architecture and design literature (Papers 1 and 2), but none of the papers describe a system that also embeds funding applications, training and mentoring within the same portal. Funding access is addressed only from a financial-inclusion perspective (Paper 4), not as an integrated application-tracking feature. Training is addressed only as a standalone app comparison (Paper 6), not as part of a broader entrepreneur support platform. Mentoring is addressed as a separate, standalone matching platform (Paper 3). Networking is addressed only through a general social media community, not a purpose-built entrepreneur networking module (Paper 5). None of the seven papers discusses personalised recommendations, secure document management, or integrated notifications as part of an entrepreneur-support system, so these features are not directly supported by the reviewed literature and are treated in this report as part of the proposed solution rather than as literature findings. Reporting and dashboards are only indirectly

supported, through the general finding that transparency improves citizen satisfaction with government digital services (Paper 2).

8. Research Gap

The existing literature provides good evidence for individual pieces of the problem described in Case Study 33. It shows that centralised, one-stop government portal architectures are recognised as important but lack a standard reference design (Paper 1); that specific, testable design features increase citizen satisfaction with government digital services (Paper 2); that digital mentor-matching can be made more effective with simple design interventions (Paper 3); that digital financial inclusion measurably increases women's entrepreneurship (Paper 4); that online social capital helps women recognise business opportunities (Paper 5); that digital-only training delivery struggles with completion compared to in-person training (Paper 6); and that service-quality dimensions such as trust and responsiveness matter for the success of women-led digital businesses (Paper 7).

What is missing from this literature is a study of these components working together as a single, integrated system for women entrepreneurs. Every reviewed paper studies exactly one function of entrepreneurial support in isolation, either government service portals, or mentoring platforms, or financial inclusion, or networking, or training apps, or general service quality, but none studies a platform that combines government scheme discovery, funding applications, training, mentoring and networking under one roof for women entrepreneurs specifically.

This gap matters because Case Study 33 describes exactly this kind of fragmentation as the core real-world problem: women entrepreneurs must currently visit multiple separate websites for schemes, funding, training and mentoring, which increases effort and the risk of missed opportunities. The literature confirms that each of these individual services can work well digitally, but it does not show how to combine them without losing the specific design strengths identified in each paper, such as the personalisation and transparency needed for schemes (Paper 2), the completion challenges of digital-only training (Paper 6), and the trust and responsiveness needed for user confidence (Paper 7).

The Women Entrepreneurship Support Portal is positioned to address this specific gap. Rather than proposing a new isolated tool, it combines the design lessons from all seven papers, centralised one-stop scheme discovery (Paper 1), citizen-satisfaction-driven design features (Paper 2), mentor outreach features informed by field-tested interventions (Paper 3), direct digital access to funding information (Paper 4), a structured networking and community

module (Paper 5), a training module that is aware of the completion risks of digital-only delivery (Paper 6), and attention to responsiveness, trust and empathy in how the platform is built and communicated (Paper 7), into one integrated platform for women entrepreneurs, government officers and mentors.

9. Future Research Directions

Based on the research gap identified above, the following future directions are suggested for the Women Entrepreneurship Support Portal and for further study:

- Better centralised scheme information: continue developing a single searchable index of government schemes, following the one-stop portal building blocks identified in Paper 1.
- Personalised scheme recommendations: use an entrepreneur's business profile to surface the most relevant schemes first, building on the personalisation factor shown to increase satisfaction in Paper 2.
- Improved funding opportunity discovery and tracking: extend the funding module so entrepreneurs can track the status of an application after submission, reflecting the transparency and self-service factors from Paper 2 and the financing-access findings of Paper 4.
- Better mentor matching: add features such as short success-story clips or examples of prior mentor-mentee interactions on mentor profiles, following the effective intervention tested in Paper 3.
- Training programme design that accounts for completion risk: combine digital training content with scheduled sessions, reminders, or optional in-person or live-online components, in direct response to the low completion rates for app-only training found in Paper 6.
- Networking and community features: provide structured spaces (such as discussion boards or interest groups) for women entrepreneurs to build the kind of bridging social capital shown to be useful in Paper 5.
- Automated notifications: keep entrepreneurs informed about scheme deadlines, application status changes, and upcoming training or networking events.
- Secure document management: continue to support secure storage and verification of business and financial documents needed for scheme and funding applications.

- Better dashboards and reports: extend reporting for government officers, building on the transparency and accuracy factors shown to matter in Paper 2.
- Attention to responsiveness, trust and empathy: monitor and improve how quickly the platform responds to entrepreneur queries and applications, following the service-quality dimensions found to matter in Paper 7.

These directions are kept realistic for a student project and do not introduce artificial intelligence, blockchain, or other advanced technologies that are not supported by the reviewed research.

10. Relevance to Women Entrepreneurship Support Portal

The findings from the literature survey connect to the Women Entrepreneurship Support Portal through the following flow: Existing Research -> Existing Limitations -> Research Gap -> Women Entrepreneurship Support Portal -> Expected Improvement.

Existing research (Papers 1 to 7) shows that individual digital tools for government services, mentoring, funding access, networking, training and service quality can each work well when properly designed. Existing limitations show that these tools are studied and built separately, that digital-only training faces completion challenges, and that service-quality gaps remain even on digital platforms. The research gap is the absence of a single, integrated platform that brings these separate functions together specifically for women entrepreneurs. The Women Entrepreneurship Support Portal is the proposed response to this gap, and the expected improvement is a single platform where a woman entrepreneur can register, discover schemes, apply for funding, join training, connect with mentors, and network with peers, instead of navigating several disconnected systems.

Specifically, the literature supports the features already identified in the previous project report as follows: entrepreneur registration and business profile are supported by the personalisation findings in Paper 2; government scheme discovery and eligibility checking are supported by the one-stop portal architecture in Paper 1 and the design factors in Paper 2; funding opportunities and applications are supported by the digital financial inclusion findings in Paper 4; training programmes are informed, and made more realistic, by the completion-rate findings in Paper 6; mentoring is supported directly by the field experiment in Paper 3; networking and community activities are supported by the social capital findings in Paper 5; and document management, notifications, and reports/dashboards are supported indirectly by the

transparency, accuracy and self-service factors identified in Paper 2, together with the trust and responsiveness factors identified in Paper 7.

11. Proposed Software Engineering Solution

Based on the case study and the literature survey, the proposed solution is the Women Entrepreneurship Support Portal already defined in the previous project report: a centralised, role-based digital platform, built with a React/Vite frontend, a Node.js/Express backend, and a SQLite-based database, deployed using Docker. This report does not redesign that system; it uses the literature findings to justify and refine the existing feature set.

Through the portal, a woman entrepreneur can register and log in, create and maintain a business profile, search for and check eligibility for government schemes, apply for funding opportunities, enrol in training programmes, connect with mentors, participate in networking and community activities, manage supporting documents, receive notifications about deadlines and updates, and track her own applications and activities through a personal dashboard.

Government officers and other authorised users can manage scheme listings, review and process funding and scheme applications, verify business/MSME details, and view administrative reports and dashboards summarising platform activity. Mentors and trainers can manage their own profiles, respond to mentorship requests, and manage training programme content and sessions.

This proposed solution keeps the system simple and student-appropriate, while directly addressing the fragmentation problem in Case Study 33 and reflecting the design lessons drawn from the seven reviewed papers, particularly the need for a centralised architecture, transparent and self-service design, realistic training delivery, and attention to trust and responsiveness.

12. Conclusion

This literature survey reviewed seven peer-reviewed research papers directly related to the problems addressed by the Women Entrepreneurship Support Portal: government portal architecture, e-government service design, online mentoring, digital financial inclusion, online social capital and networking, app-based entrepreneurship training, and service quality on digital platforms used by women entrepreneurs.

The common approach across the papers is the use of real-world data, whether from platform logs, national surveys, randomised trials, or structured reviews, to test how digital tools actually affect entrepreneurs, rather than relying only on theory. A common limitation across the papers is that each one studies a single support function in isolation, and several are limited to a specific country, platform, or sample.

The research gap identified in this report is the lack of a single, integrated platform that combines government scheme discovery, funding, training, mentoring and networking for women entrepreneurs. Future directions include personalised scheme recommendations, application tracking, better mentor matching, more realistic training delivery, structured networking features, and continued attention to service quality, trust and responsiveness.

Overall, the Women Entrepreneurship Support Portal addresses the problem identified in Case Study 33 by combining, in one centralised system, the individual design lessons drawn from the reviewed literature, offering women entrepreneurs a single place to discover and use the government support that is available to them.

13. References

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Rubrics for Calculation

Evaluation Criteria	Excellent (4 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Selection & Review of Research Papers(25%)	Selects highly relevant, recent, credible papers and provides comprehensive reviews of objectives, methodology, tools, datasets, and findings	Selects relevant papers and provides good reviews with minor gaps	Selects moderately relevant papers with basic reviews	Papers are irrelevant or reviews are incomplete
Comparative Analysis(20%)	Provides a detailed comparison of methodologies, tools, datasets, results, and performance using a clear comparison table	Provides a good comparison with relevant parameters	Provides limited comparison with few parameters	Little or no meaningful comparison
Critical Analysis & Research Gap(25%)	Clearly analyzes limitations and identifies significant, well-supported research gaps	Identifies relevant limitations and research gaps	Identifies basic gaps with limited analysis	Research gaps are unclear or missing
Future Directions & Relevance(15%)	Proposes practical and innovative future directions directly relevant to the assigned problem statement	Proposes relevant future directions	Provides general future suggestions	Future directions are missing or unrelated
Documentation , Citations & References(15%)	Excellent academic structure, clear presentation, accurate citations, and consistent referencing	Well-organized with minor formatting/citation issues	Adequately documented with some errors	Poorly organized with inadequate or incorrect references

Criteria	Mark
Selection & Review of Research Papers(25%)	/5
Comparative Analysis(20%)	/5
Critical Analysis & Research Gap(25%)	/5

Future Directions & Relevance(15%)	/5
Documentation, Citations & References(15%)	/5
TOTAL	/25